Real-World Adaptation and Enhancement of Recurrent Marginal Structural Networks to Estimate Causal Impact of Sales and Account Management Teams

Zhiyuan Shen **Shuo Yang** **Ashish Patira** **Hossein Keshavarz**
Michelle Song **Saad Zaheer** **Palash Siddamsettiwar** **Joey Kao** **Nicole Cheng**
Emmanuel Ekwedike **Matthew Sisco** **Lei Kang** **Khalifeh AlJadda**

Google

Abstract

In industrial settings like Google Customer Solutions (GCS), estimating the incremental value of human and agentic-AI sales interventions is challenging because treatments are dynamic and sequential, and their assignment is influenced by complex time-varying confounders that also affect advertiser outcomes. To address these challenges, we introduce several methodological extensions to RMSN: attention-augmented sequence modeling with layer normalization, long-horizon weight stabilization, multi-action counterfactual prediction with scalable Shapley attribution, hyperparameter ensembling, and explicit handling of structurally untreated control accounts. The resulting framework supports scalable estimation and attribution of longitudinal treatment effects across heterogeneous sales programs. We demonstrate improved multi-step prediction performance over the original RMSN baseline on two public benchmark datasets.

1 Introduction

Sales teams and AI agents on large online platforms interact repeatedly with advertisers through account reviews, targeted recommendations, and specialized programs. Estimating the incremental revenue caused by these engagements is important for resource allocation, incentive design, and long-term business planning, but is not a standard attribution problem. Seller engagements are dynamic, sequential, and heterogeneous; time-varying confounders, such as evolving advertiser budgets, market trends, policy changes, and internal platform optimizations simultaneously influence both seller behavior (treatment assignment) and advertiser spend (outcome); the underlying dataset is of immense scale and structural complexity, spanning high-dimensional covariates, and long horizons.

Building on RMSN [Lim et al., 2018], which uses recurrent propensity networks to estimate inverse probability of treatment weights (IPTWs) and trains a sequence-to-sequence outcome model with those weights to adjust for time-varying confounding, we develop an industrial-scale framework for longitudinal sales-impact estimation. Applying RMSN to GCS requires addressing challenges beyond those emphasized in benchmark settings: millions of high-dimensional trajectories, 13–52-week forecasting horizons, multiple concurrent interventions, unstable cumulative inverse-propensity weights, highly imbalanced program sizes, and structurally untreated control accounts.

Our contribution is an industrial adaptation of RMSN that addresses these constraints while preserving explicit propensity-based adjustment for time-varying confounding. We augment sequence modeling with causal attention and layer normalization; stabilize long-horizon estimation through decoder weight capping and model ensembling; and support concurrent treatments with Shapley-based

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

attribution. We demonstrate improved multi-step prediction performance over the original RMSN baseline on two public benchmark datasets.

2 Related Work

Marginal Structural Models (MSMs) use inverse probability of treatment weighting (IPTW) to adjust for measured time-varying confounders [Robins et al., 2000]. RMSN [Lim et al., 2018] parameterizes propensity and outcome models with recurrent networks, enabling nonlinear, history-dependent counterfactual forecasting. Alternative neural approaches learn treatment-balanced representations, as in the Counterfactual Recurrent Network [Bica et al., 2020], or use attention-based sequence models, as in the Causal Transformer [Melnychuk et al., 2022]. Our goal is complementary: we retain the explicit weighting mechanism of RMSN while adapting its architecture and training procedure to long-horizon, multi-action industrial data, where stability, scalability, and attribution are first-order requirements.

3 Methodology

3.1 RMSN formulation

For advertiser i at time t, let $Y_{t,i}$ be the outcome, $A_{t,i}$ the intervention vector, $L_{t,i}$ time-varying covariates, and X_i advertiser-specific static features. Let $\bar{H}_{t,i} = (\bar{L}_{t,i}, \bar{A}_{t-1,i}, X_i)$ denote observed history of advertiser i. RMSN first fits denominator and numerator propensity networks and forms the stabilized per-step weight [Robins et al., 2000]

$$sw_{t,i} = \frac{\hat{p}_{\text{num}}(A_{t,i} \mid \bar{A}_{t-1,i})}{\hat{p}_{\text{den}}(A_{t,i} \mid \bar{L}_{t,i}, \bar{A}_{t-1,i}, X_i)}. \tag{1}$$

The denominator models treatment selection probability given the full history of confounders and past actions, whereas the numerator models a marginal adoption process that depends only on past actions to stabilize the resulting weights. The resulting sequence of weights $(sw_{1,i}, \ldots, sw_{T_i,i})$ is used during encoder and decoder training to adjust for measured time-varying confounding when estimating longitudinal causal effects.

An LSTM encoder learns a propensity-weighted representation of each advertiser's evolving trajectory by reweighting the training examples using the stabilized weights computed by Equation (1):

$$\mathcal{L}_{\text{enc}} = \sum_{i=1}^{I} \sum_{t=1}^{T_i} sw_{t,i}\, \ell\Big(Y_{t+1,i}, \hat{Y}_{t+1,i}\Big), \tag{2}$$

The decoder parameters are learned by minimizing a stabilized-weighted empirical risk over all advertisers, start times, and prediction horizons, using a per-step loss ℓ.

$$\mathcal{L}_{\text{dec}} = \sum_{i=1}^{I} \sum_{t=1}^{T_i} \sum_{\tau=2}^{\min(T_i - t, \tau_{\max})} sw_{t,\tau,i}\, \ell\Big(Y_{t+\tau,i}, \hat{Y}_{t+\tau,i}\Big), \tag{3}$$

where $sw_{t,\tau,i} = \prod_{u=t}^{t+\tau} sw_{u,i}$ is the cumulative stabilized weight and $\tau_{\max}$ is the maximum decoder horizon. These objectives show that propensity weighting adjusts both one-step encoder learning and multi-step decoder forecasting. This adjustment assumes that all relevant time-varying confounders are observed and that each evaluated treatment has sufficient support given the observed history.

3.2 Methodological Extensions to RMSN for the GCS Setting

Direct application of RMSN to GCS requires adaptations for longer horizons, concurrent interventions, heterogeneous program sizes, and structurally constrained treatment eligibility. We retain the RMSN identification strategy while introducing the following methodological extensions.

Multi-Head Attention and Layer Normalization The original RMSN architecture [Lim et al., 2018] relies on fixed-dimensional recurrent states to summarize the preceding sequence at each prediction step. To reduce this dependence, we apply causal multi-head self-attention to the time-indexed

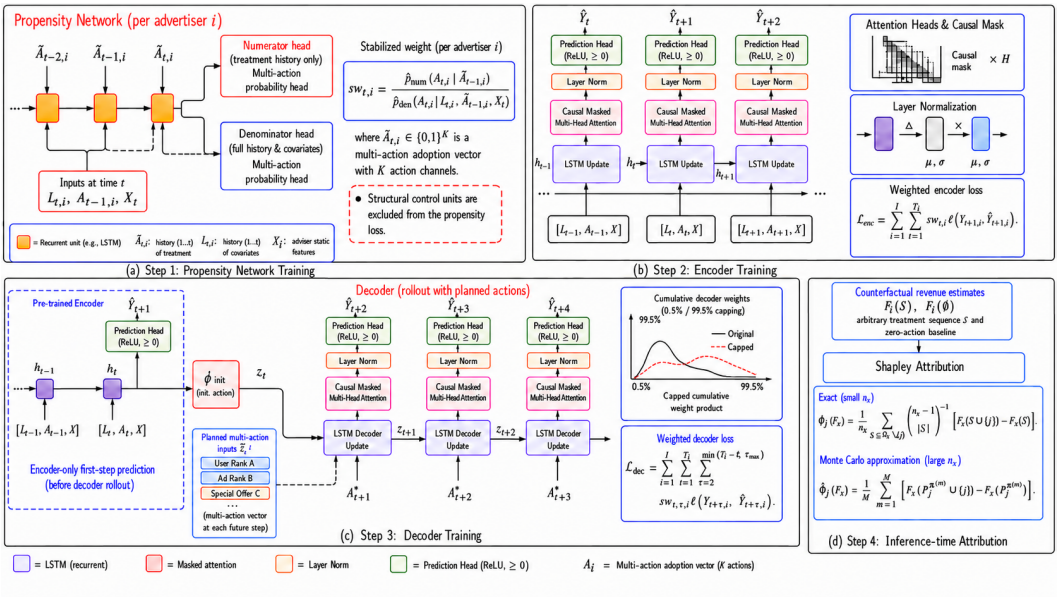


Figure 1: RMSN-based framework. Propensity networks construct stabilized weights; an encoder summarizes reweighted history; and the decoder forecasts outcomes under specified future intervention sequences.

RNN outputs in both the encoder and decoder, enabling each prediction to selectively aggregate earlier recurrent representations and better capture long-range temporal interactions without accessing future information. We also introduce layer normalization as a tunable architectural component to stabilize training on variable-length sequences. Across measurement quarters, configurations selected by Google Vizier—a black-box optimization service for hyperparameter and system-design tuning—consistently enable layer normalization, providing empirical evidence of its benefit.

Multi-Action Decoder and Shapley-Based Attribution Because multiple sellers or programs may engage the same advertiser concurrently, we extend the decoder input from a single treatment indicator to a multi-action vector, allowing concurrent interventions at each time step. The decoder predicts revenue trajectories under arbitrary multi-action sequences, and we estimate the joint impact of a given sequence as the difference between its predicted revenue trajectory and that under the corresponding zero-action sequence.

We attribute this combined impact using Shapley values, defined as each intervention's expected marginal contribution over uniformly random hypothetical intervention orderings. Because exact enumeration is exponential in the number of interventions, we approximate the attribution using M sampled permutations $\{\pi^{(m)}\}_{m=1}^{M}$:

$$\widehat{\Phi}_j(F_x) = \frac{1}{M}\sum_{m=1}^{M}\left[F_x(P_j^{\pi^{(m)}} \cup \{j\}) - F_x(P_j^{\pi^{(m)}})\right], \tag{4}$$

where $P_j^{\pi^{(m)}}$ denotes the set of interventions that precede j in $\pi^{(m)}$, $F_x(S)$ represents the predicted revenue for advertiser x when treated by the intervention set S. This decomposition allocates shared and synergistic lift among concurrent interventions. For an advertiser x with adopted intervention set $\mathcal{O}_x = \{1, \ldots, n_x\}$, exact computation requires $(O(2^{n_x}))$ value-function evaluations whereas the Monte Carlo approximation reduces the computational complexity to $(O(Mn_x))$. In practice, we use exact attribution for small treatment sets while using a Monte Carlo estimator for larger intervention sets with M chosen to satisfy a fixed computational budget.

Weight Stabilization and Hyperparameter Ensembling To stabilize long-horizon estimates, we clip cumulative decoder IPTWs to the 0.5th–99.5th percentile range while leaving encoder weights

uncapped. We tune hyperparameters using a 50-trial adaptive Vizier search; in production, we additionally average predictions from the $N = 4$ models with the lowest validation losses to reduce sensitivity to initialization and stochastic optimization, with N selected empirically.

Handling Structural Zero-Probability Treatments RMSN's IPTW identification requires treatment positivity: each evaluated treatment must have nonzero probability conditional on observed history. In GCS, quarterly control accounts are prohibited from receiving seller engagements, giving them structurally zero treatment probability. Including these accounts in propensity estimation forces the model to fit a degenerate assignment process and empirically degrades post-weighting covariate balance. We therefore exclude them from the propensity loss while retaining their untreated trajectories in outcome-model training.

4 Experiment

Public benchmarks We evaluate on two datasets constructed by extracting/generating 117-step sequences. The **Starbucks Rewards** benchmark [Chen, 2020] simulates customer responses to 10 promotional offers (BOGO, discount, and informational), with demographics and offer metadata as covariates and transactions as the outcome. The generated dataset contains 16,788 users, 97.53% of whom have at least one recorded transaction. The **Tumor Growth** simulator used by the original RMSN paper [Lim et al., 2018] models chemotherapy and radiotherapy under time-varying patient state. The generated dataset contains 10,000 patients, with a 3.38% of outcome observations are zero.

Evaluation protocol We evaluate the adapted predictor against the original RMSN on the Tumor Growth and Starbucks Rewards benchmarks using 10-fold cross-validation and a 13-step forecasting horizon, where the cross-validation splits are constructed at the patient/user trajectory level.

For models trained with a Box–Cox-transformed target, we inverse-transform the predictions before evaluation. Thus, both approaches are evaluated against the observed outcomes in their original scale. Following [Lim et al., 2018], we normalize the MAE at each prediction horizon by the simulator's predefined maximum tumor volume, ($V_{\max} = 1150\text{cm}^3$), and report ($100 \times \text{MAE}/V_{\max}$) as a percentage. This dimensionless metric expresses prediction error relative to a fixed, clinically meaningful upper bound. Similarly, we normalize the MAE by the maximum transaction amount in the Starbucks dataset, 1001.85. As Table 1 shows, across the 10 cross-validation folds, on Tumor Growth, our adapted RMSN reduces overall NMAE from ($7.28\% \pm 1.235\%$) to ($0.32\% \pm 0.072\%$). On Starbucks Rewards, overall NMAE decreases from ($0.150\% \pm 0.007\%$) to ($0.0040\% \pm 0.00008\%$).

Table 1: Normalized MAE (%) by forecast step over 10 cross-validation folds.

Dataset	Model	Statistic	1	2	3	4	5	6	7	8	9	10	11	12	13	Overall
Tumor	RMSN	Mean	4.92	5.07	5.14	5.35	5.84	6.55	7.23	7.84	8.43	8.96	9.39	9.74	10.11	7.28
		Std.	1.521	0.918	0.404	0.522	0.761	1.068	1.436	1.726	1.903	2.000	2.024	1.973	1.932	1.235
	Adapted RMSN	Mean	0.16	0.14	0.18	0.21	0.25	0.28	0.31	0.35	0.38	0.41	0.45	0.49	0.53	0.32
		Std.	0.052	0.049	0.038	0.048	0.048	0.055	0.064	0.075	0.087	0.097	0.109	0.121	0.136	0.072
Starbucks	RMSN	Mean	0.19	0.03	0.03	0.03	0.05	0.03	0.18	0.04	0.04	0.04	0.05	0.04	1.20	0.15
		Std.	0.030	0.003	0.002	0.002	0.004	0.002	0.021	0.002	0.002	0.003	0.005	0.003	0.078	0.007
	Adapted RMSN	Mean	0.0040	0.0038	0.0038	0.0038	0.0039	0.0038	0.0040	0.0038	0.0038	0.0038	0.0039	0.0038	0.0052	0.0040
		Std.	0.00008	0.00007	0.00007	0.00007	0.00007	0.00007	0.00008	0.00008	0.00008	0.00008	0.00008	0.00008	0.00009	0.00008

5 Conclusion

We presented an industrial adaptation of RMSN for estimating and attributing the effects of longitudinal, potentially concurrent sales interventions. The framework preserves propensity-based adjustment for measured time-varying confounding while adding causally masked attention and layer normalization, multi-action counterfactual forecasting with scalable Shapley attribution, hyperparameter ensembling, and treatment-eligibility-aware propensity training. These adaptations target practical challenges posed by long trajectories, overlapping programs, estimate variability, and structurally untreated control accounts.

The results on public benchmarks demonstrate improved multi-step forecasting accuracy and stability, supporting the prediction-oriented adaptations. Together with the production-specific extensions for attribution, ensembling, and treatment eligibility, the resulting framework addresses key requirements of scalable longitudinal treatment-effect estimation in industrial settings.

References

Ioana Bica, Ahmed M. Alaa, James Jordon, and Mihaela van der Schaar. Estimating counterfactual treatment outcomes over time through adversarially balanced representations. In *International Conference on Learning Representations (ICLR)*, 2020.

Linda Chen. Starbucks capstone challenge dataset. `https://github.com/linnndachen/starbucks_capstone`, 2020.

Bryan Lim, Ahmed M Alaa, and Mihaela van der Schaar. Forecasting treatment responses over time using recurrent marginal structural networks. In *Advances in Neural Information Processing Systems*, volume 31, pages 7483–7493, 2018.

Valentyn Melnychuk, Dennis Frauen, and Stefan Feuerriegel. Causal transformer for estimating counterfactual outcomes. In *International Conference on Machine Learning (ICML)*, 2022.

James M Robins, Miguel Á Hernán, and Babette A Brumback. Marginal structural models and causal inference in epidemiology. *Epidemiology*, 11(5):550–560, 2000.